

Frequency-Aware Dual-Branch LIIF for Super-Resolution

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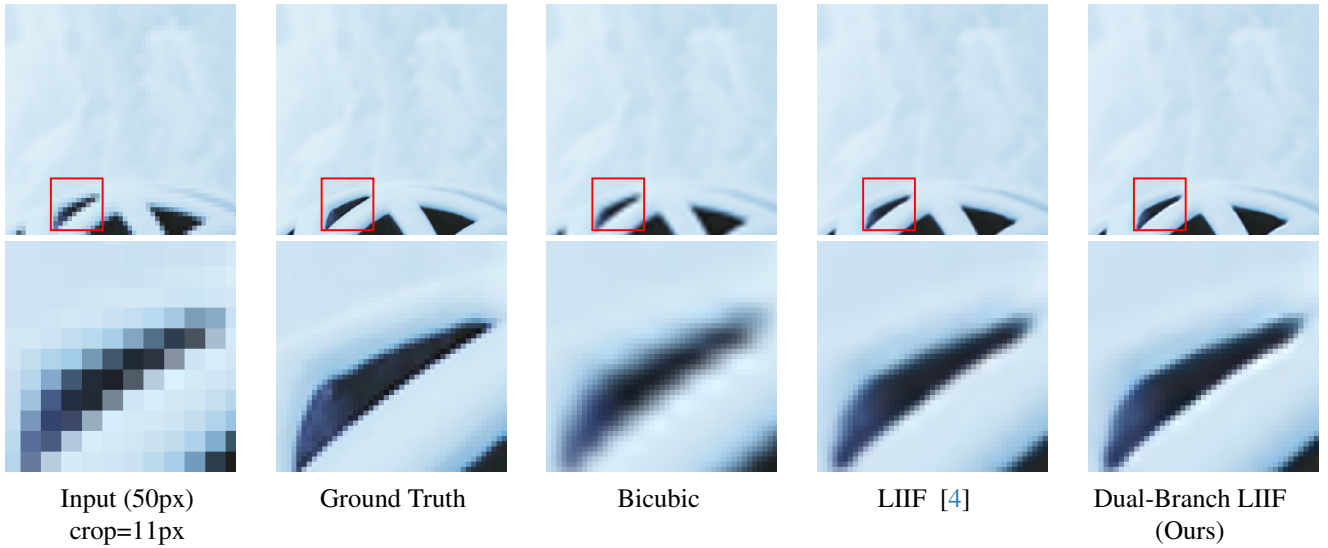


Figure 1. **Qualitative comparison of arbitrary-scale super-resolution ($\times 4$).** The input is a 50×50 patch extracted from an unseen image in the DIV2K validation set. The red bounding boxes indicate the 11×11 cropped regions used for detailed demonstration. Compared to the baseline LIIF and bicubic interpolation, our proposed Dual-Branch LIIF successfully restores high-frequency details without over-smoothing, explicitly guided by the decoupled high-frequency residual map.

Abstract

Implicit Neural Representation (INR) has recently emerged as a highly effective paradigm for arbitrary-scale image super-resolution (SR). Specifically, the Local Implicit Image Function (LIIF) achieves continuous image representation by mapping 2D spatial coordinates to RGB values via a multi-layer perceptron (MLP), enabling infinite-resolution upsampling. However, despite its capabilities, LIIF suffers from the spectral bias inherent to coordinate-based MLPs. As a result, the network prioritizes learning low-frequency structural components than learning high-frequency details, leading to over-smoothed edges and a loss of fine-grained textures. To mitigate this limitation without introducing architectural complexity, we propose a data-centric method: Dual-Branch LIIF. We train a vanilla LIIF model to reconstruct the base of the image, while simultaneously training an additional LIIF model specialized

in high-frequency details by using a high-frequency residual dataset. When given a SR task, the predictions from both branches are linearly combined using a scaling factor. Experiments demonstrate that this straightforward summation successfully restores sharp, high-fidelity details, yielding improvements in PSNR and SSIM scores when compared to the baseline LIIF.

1. Introduction

Recently, there have been numerous approaches to overcome resolution constraints in image upsampling. Implicit Neural Representations (INRs) stand out when addressing this challenge. They overcome the inherent limitations of the discrete digital pixels by parameterizing images as continuous functions, mapping spatial coordinates to their corresponding RGB values. As a type of INR, the Local Im-

PLICIT Image Function (LIIF) has demonstrated remarkable capability in upsampling images to unseen scales without losing fidelity through its continuous formulation.

Despite its capabilities, the Multi-Layer Perceptron (MLP) used as its decoder inherently suffers from spectral bias. As demonstrated by Rahaman et al. [11], MLPs that use Rectified Linear Unit (ReLU) as their activation function tend to rapidly learn the low-frequency components of the image, while treating the high-frequency details as noise. Further developing this theory, Tancik et al. [13] mathematically proved through Neural Tangent Kernel (NTK) analysis that standard MLPs mapping low-dimensional coordinates experience rapid eigenvalue decay in the high-frequency spectrum. Consequently, this mathematical limitation manifests as a severe optimization bottleneck during training, fundamentally restricting the network’s ability to restore high-frequency details and resulting in over-smoothed outputs.

Therefore, to overcome this bottleneck for high-frequency details, we propose a data-centric approach rather than manipulating the mathematical structure of the decoder. In our Dual-Branch method, alongside the original dataset, we prepare a new dataset that exclusively contains the high-frequency details. Specifically, we apply a low-pass filter to the image and subtract the filtered output from the original image to obtain a high-frequency residual map. We then train two separate LIIF models on their respective datasets. Training two distinct models on different dataset allows each network to focus solely on its target frequency domain. This decoupled high-frequency-specialized model alleviates the burden on the baseline LIIF model in restoring high-frequency details. During inference, the two models generate respective predictions, and the high-frequency output is then added to the base output after being multiplied by a fixed scaling factor. The use of this scaling factor suppresses overshooting noise and allows a well-balanced enhancement in PSNR metrics. Since our dual-branch pipeline does not introduce any architectural modifications, it holds strong potential to be generalized and be readily applied to other INRs.

The main contributions of our work are summarized as follows. **Contribution 1: Novel Methodology** We introduce a data-centric Dual-Branch framework that separates the learning process of low-frequency and high-frequency components. This approach successfully overcomes the spectral bias of the baseline LIIF model without making any architectural modifications. **Contribution 2: Alleviate Burden and Improve Performance** We prove through experiments that utilizing an additional high-frequency specialized model significantly reduces the burden of the baseline model to restore high-frequency details. As a result, our pipeline successfully reconstructs detailed edges that were previously over-smoothed, outperforming the original LIIF

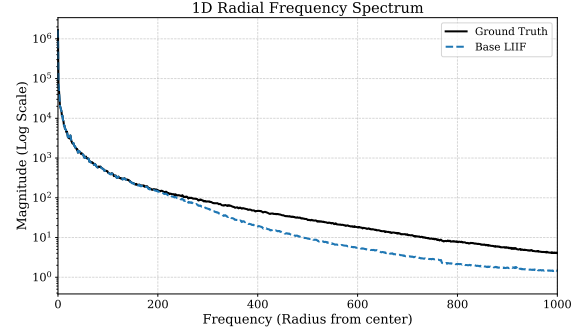


Figure 2. 1D radial frequency spectrum analysis of the ground truth and the baseline LIIF prediction. The baseline model exhibits a clear spectral bias, showing a magnitude drop in the high-frequency regions compared to the ground truth.

model in arbitrary-scale super-resolution.

2. Related Work

2.1. Arbitrary-Scale Image Super-Resolution

Traditional deep learning-based super-resolution (SR) methods rely on fixed-scale upsampling modules, which necessitate training separate models for distinct integer scales (e.g., $\times 2$, $\times 3$, $\times 4$). To overcome this inconvenience, MetaSR [5] first introduced the concept of arbitrary-scale SR by dynamically predicting the weights of upsampling filters using a meta-network. Later, LIIF [4] significantly advanced the field by formulating images as continuous functions parameterized by MLPs, directly decoding RGB values from continuous coordinates and neighboring feature vectors. However, while LIIF achieves infinite-resolution extrapolation, it often produces over-smoothed textures. To tackle this, numerous approaches have been proposed, including LTE [8], which incorporates a Local Texture Estimator to guide the network using frequency-domain information. While effective, transforming features into the Fourier domain and estimating amplitude and phase introduces considerable architectural complexity. In contrast, our approach directly isolates and injects high-frequency residuals entirely within the spatial domain. This dual-branch strategy avoids complex frequency transformations, providing a much more direct, intuitive, and effective solution for preserving fine details in continuous image representations.

2.2. Spectral Bias in Neural Networks

The tendency of neural networks to prioritize learning low-frequency signals while struggling to capture high-frequency details is a well-documented phenomenon known as spectral bias [11]. It mathematically explains why standard MLP-based implicit functions, such as the original LIIF, generate blurry edges and fail to reconstruct sharp

textures. We can prove the presence of this bias in Figure 2. To mitigate this, several studies have attempted modifying the input mapping or activation functions. For instance, Tancik et al. [13] demonstrated that mapping input coordinates using Fourier features allows MLPs to better learn high-frequency functions. Similarly, SIREN [12] replaced the standard ReLU with periodic sine activation functions to capture fine details in implicitly represented scenes. While these approaches enhance representational capacity, altering the fundamental activation or coordinate space can sometimes compromise the smooth continuous nature required for natural image interpolation. Instead of modifying the internal non-linearities, our method addresses the spectral bias by explicitly feeding an isolated high-frequency residual map to a dedicated network branch, enforcing the network to learn the missing details directly.

2.3. Frequency-Decoupled Architectures

In the broader context of computer vision, decoupling low-frequency and high-frequency components has been a proven strategy to improve both efficiency and representation quality. Classical architectures like Laplacian Pyramid Networks (LapSRN) [7] progressively predict high-frequency residuals at multiple scales to reconstruct sharp images. Furthermore, Octave Convolution (Oct-Conv) [3] factorizes feature maps into spatial frequency domains, processing them at different resolutions to reduce redundancy while maintaining representation power. These methods successfully demonstrate that guiding a network with isolated frequency bands yields superior edge reconstruction. However, such spatial decoupling mechanisms have been primarily designed for discrete convolutional neural networks (CNNs) and remain underexplored in continuous implicit neural representations. Inspired by this well-established philosophy, our Dual-Branch LIIF pioneers the integration of a frequency-decoupled architecture into the coordinate-based implicit function, seamlessly blending the strengths of structural spatial filtering with infinite-resolution rendering.

3. Method

3.1. Overview

The overall architecture of our proposed Dual-Branch LIIF is illustrated in Figure 3. To explicitly guide the network in restoring sharp textures without losing structural continuity, we decouple the super-resolution process into two parallel branches. For the base branch, we utilize the standard LIIF model (f_{base}) without any architectural modifications, taking the original low-resolution image as input. For the high-frequency branch (f_{hf}), we extract a high-frequency residual map by subtracting a low-pass filtered version from the original input. Both branches independently process

their respective inputs to predict high-resolution continuous representations, yielding intermediate predictions y_{base} and y_{hf} , respectively. Finally, the high-frequency prediction y_{hf} is multiplied by a scaling factor α and fused with the base prediction y_{base} to produce the final high-fidelity output y_{final} . This explicit spatial decoupling ensures that the high-frequency branch strictly focuses on detail recovery, effectively mitigating the spectral bias of the base function.

3.2. Background: Local Implicit Image Function

Local Implicit Image Function (LIIF) [4] defines an image as a continuous representation in a 2D domain. Given a low-resolution (LR) image I_{LR} , an encoder E_φ first extracts a 2D feature map $z = E_\varphi(I_{LR})$, where each pixel in the feature map is treated as a latent code distributed in the continuous spatial domain. To predict the RGB value s at an arbitrary coordinate x , LIIF utilizes a decoding function f_θ , typically parameterized as a multi-layer perceptron (MLP). The decoding process is formulated as:

$$s(x) = \sum_{j \in \mathcal{J}} w_j \cdot f_\theta(z_j, x - x_j) \quad (1)$$

where $\mathcal{J}$ denotes the indices of the four nearest latent codes around the coordinate x , and z_j is the j -th nearest latent feature vector located at x_j . The term $x - x_j$ represents the relative coordinate, and w_j are the bilinear interpolation weights (local ensemble weights) that ensure spatial continuity across the reconstructed image. While this coordinate-based representation enables infinite-resolution extrapolation, a standalone MLP with ReLU activations inherently prioritizes the reconstruction of low-frequency components, a phenomenon known as spectral bias [11]. This leads to the loss of fine details in the high-resolution output. Our proposed method addresses this fundamental limitation by explicitly isolating and processing the high-frequency components through a dedicated branch, which we describe in the following sections.

3.3. High-Frequency Separation in Spatial Domain

In this section, we detail the explicit extraction of the high-frequency residual map. Unlike previous approaches that rely on complex frequency-domain transformations such as Fourier or Wavelet transforms, we perform decoupling directly in the spatial domain for both computational efficiency and architectural simplicity. Given a low-resolution (LR) image I_{LR} , we first apply a Gaussian blur filter G_σ with a standard deviation σ to obtain the low-frequency base component:

$$I_{base} = G_\sigma(I_{LR}) \quad (2)$$

The high-frequency residual map I_{hf} is then extracted by calculating the element-wise difference between the origi-

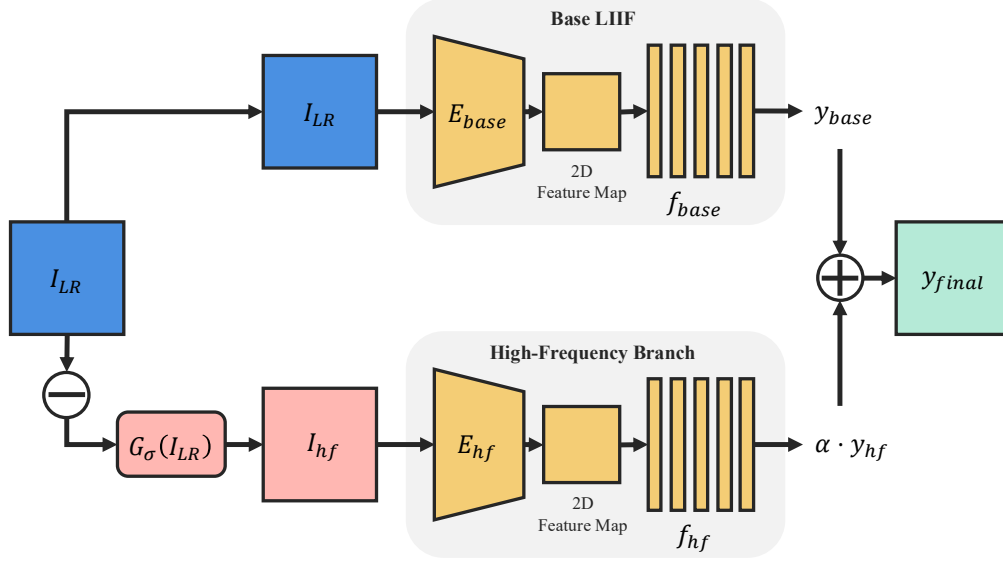


Figure 3. Overview of the proposed Dual-Branch LIIF architecture. The low-resolution input image (I_{LR}) is sent to two parallel pipelines. In the Base LIIF branch, the original I_{LR} is passed through an encoder (E_{base}) to extract a 2D feature map, which is then decoded by a base implicit function (f_{base}) to predict the base continuous representation (y_{base}). Concurrently, the High-Frequency Branch extracts a high-frequency residual map (I_{hf}) using a Gaussian filter (G_{σ}). This residual is processed by a separate encoder (E_{hf}) and a high-frequency implicit function (f_{hf}). Both of our branches use EDSR[9] as their encoder. Finally, the high-frequency prediction is multiplied by a scaling factor α ($\alpha \cdot y_{hf}$) and fused with y_{base} via element-wise addition to synthesize the final high-resolution output (y_{final}).

nal input and its blurred counterpart:

$$I_{hf} = I_{LR} - I_{base} \quad (3)$$

This simple yet effective spatial filtering allows the network to bypass the complex estimation of phase and amplitude required in frequency-domain approaches. By providing I_{hf} as a separate input, we enforce a strong structural prior that guides the subsequent branch to focus exclusively on capturing sharp transitions and fine textures, which are typically lost due to the spectral bias of standard MLPs.

3.4. Dual-Branch Implicit Representation and Fusion

Our proposed framework processes the decoupled inputs through two parallel and independent branches. Each branch consists of a dedicated encoder (E_{base}, E_{hf}) and a coordinate-based MLP decoder (f_{base}, f_{hf}). Specifically, the base branch reconstructs the underlying structure from I_{base} , while the high-frequency branch is specialized to learn the residual details from I_{hf} . For a target coordinate x , the intermediate predictions from each branch are generated as follows:

$$y_{base} = f_{base}(z_{base}, x, c) \quad (4)$$

$$y_{hf} = f_{hf}(z_{hf}, x, c) \quad (5)$$

where z_{base} and z_{hf} are the neighboring latent vectors extracted by the respective encoders, and c denotes the cell size representing the target resolution. The final high-resolution output y_{final} is synthesized by fusing these predictions. To maximize the reconstruction performance, we identify and employ an empirically determined optimal scaling factor α , which is applied to the high-frequency prediction to calibrate the influence of recovered details before the final fusion:

$$y_{final} = y_{base} + \alpha \cdot y_{hf} \quad (6)$$

By finding the optimal balance between the structural base and the high-frequency components, this dual-branch formulation effectively mitigates the spectral bias inherent in standard MLPs, ensuring high-fidelity restoration of fine-grained textures.

4. Experiments

4.1. Experimental Setup

We use DIV2K dataset [1] in training and testing our method like the original LIIF paper [4]. Among the 1000 2K resolution images, we use predetermined 800 images in the DIV2K training set to train our models. When it comes to testing, we use unseen 100 images in DIV2K valid set to calculate PSNR scores.

Table 1. Quantitative comparison on the DIV2K validation set (PSNR in dB). The best results are highlighted in **bold**. Except for Bicubic, EDSR-baseline is used as the encoder for all methods. The results are obtained from the original manuscripts [2, 4, 8, 10, 14].

Method	In-distribution			Out-of-distribution		
	$\times 2$	$\times 3$	$\times 4$	$\times 6$	$\times 12$	$\times 18$
Bicubic	31.01	28.22	26.66	24.82	22.27	21.00
EDSR-baseline [9]	34.55	30.90	28.94	-	-	-
Meta-SR [5]	34.64	30.93	28.92	26.61	23.55	22.03
LIIF [4]	34.67	30.96	29.00	26.75	23.71	22.17
UltraSR [14]	34.69	31.02	29.05	26.81	23.75	22.21
IPE [10]	34.72	31.01	29.04	26.79	23.75	22.21
LTE [8]	34.72	31.02	29.04	26.81	23.78	22.23
CLIT [2]	34.82	31.14	29.17	26.93	23.85	22.30
LIIF (Ours)	33.39	29.91	28.07	25.97	23.13	21.71
Dual-Branch LIIF (Ours)	32.98	29.89	28.19	26.07	23.2	21.77

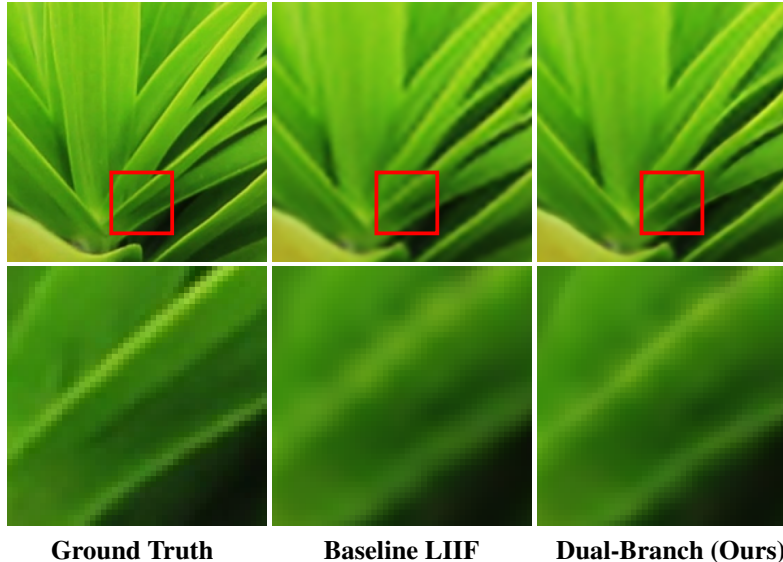


Figure 4. Visual comparison at scale factor $\times 6$. From left to right: (a) Ground Truth, (b) Baseline LIIF, and (c) our Dual-Branch LIIF.

To prepare the training data for the high-frequency (HF) branch, we extract the HF components from the ground-truth high-resolution (HR) images. Specifically, we apply a Gaussian blur filter with a kernel size of 5×5 and a standard deviation (σ) of 1.0. The high-frequency residual is then obtained by subtracting this blurred output from the original HR image.

4.2. Implementation Specifications

Since we utilize identical LIIF model from [4], we use 48×48 patches as inputs for our encoder. Then we scale the

patch with an arbitrary number between 1 and 4 before feeding our encoder. We use EDSR-baseline [9] as our encoder E_φ for both branches. To meticulously reproduce [4], we use L1 loss and the decoding function f_θ is a 5-layer MLP with ReLU activation and hidden dimensions of 256. We also use Adam [6] optimizer with an initial learning rate $1 \cdot 10^{-4}$, the learning rate decays by factor 0.5 every 200 epochs.

Though we successfully implemented model architecture and hyperparameters for our LIIF branches, we only had access to Colab pro resources for our training. Result-

ing in training time constraint of 24 hours. To enable testing within this semester we chose to keep the epoch and batch size which are 1000 and 16 respectively while not amplifying the dataset by letting the model repeat on the dataset. This modification makes our model to train 20 times less iterations than the [4], potentially implying underfitting issues in both of our branches.

4.3. Quantitative Results

In Table 1, we compare the performance of our self-trained LIIF and the proposed Dual-Branch LIIF against several off-the-shelf SR models [2, 4, 5, 8–10, 14]. Notably, our self-trained LIIF model failed to reproduce the reported PSNR scores of [4], exhibiting a substantial performance gap when compared to modern SR architectures like [2]. However, as illustrated in Figure 5, a direct comparison between our baseline LIIF and the Dual-Branch LIIF reveals consistent improvements in SSIM scores across all tested scales. Regarding PSNR, the Dual-Branch method demonstrates higher performance than the baseline specifically at scale factors greater than $\times 4$. Furthermore, as shown in Figure 6, our method effectively mitigates the spectral bias inherent in the original LIIF by successfully restoring more high-frequency details.

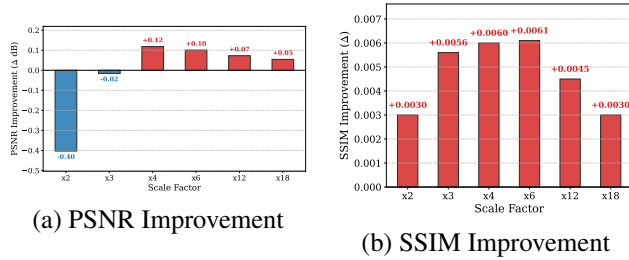


Figure 5. Performance improvements (Δ) of our proposed Dual-Branch LIIF over the baseline model across various scale factors on the DIV2K validation set. Red bars indicate positive gains, demonstrating consistent enhancements in SSIM and enhancement for scales higher than 3 in PSNR.

4.4. Qualitative Results

In Figure 4 our proposed Dual-Branch LIIF demonstrates a significant advantage over the baseline LIIF, particularly in regions with sudden color transitions. As shown in the zoomed-in views, our method reconstructs much sharper and more distinct boundaries, whereas the baseline tends to produce blurry and over-smoothed results along strong edges. Crucially, this capability to preserve high-frequency details is not limited to the in-distribution trained scales (i.e., $\times 1$ to $\times 4$). We observe that this precise reconstruction persists even when rendering at unseen scales that fall outside the trained range.

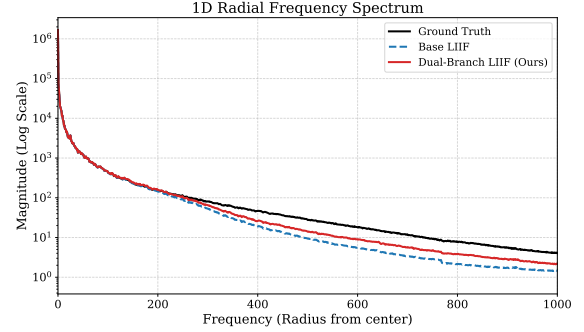


Figure 6. 1D radial frequency spectrum analysis of the ground truth, the baseline LIIF prediction and our Dual-Branch LIIF. Our method successfully reduce the magnitude drop in high-frequency field of the image.

5. Conclusion

In this paper, we presented a data-centric approach to improve the performance of the LIIF model. To overcome the inherent limitations of the MLP decoder, we trained an additional LIIF model specifically on high-frequency components extracted from the training data. By fusing a scaled restoration of this specialized model with the results from the baseline LIIF, we achieved final outputs that are superior from both quantitative and qualitative perspectives. For models that utilize MLP decoders, especially when the size of the dataset is limited, our method is expected to effectively boost performance.

We acknowledge that training our two branches for a more extensive amount of time would be ideal to fully reproduce the state-of-the-art performance and further prove our method’s capabilities. Nevertheless, we believe our dual-branch framework provides a meaningful stepping stone for addressing the spectral bias in continuous image representation.

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